10-30 第1篇

阅读理解（说明文） Why Does the Moon Turn Red During a Lunar Eclipse?

① On the evening of **20 March 2027**, thousands of people in East Asia went outside their flats and looked up. The full moon slowly walked into the earth's shadow. But instead of disappearing into black, it **turned a deep, soft red**, the colour of old cherries. Many children asked their parents the same question: "Mum, why is the moon bleeding?" The moon, of course, was not bleeding. To answer the children, we must first understand three small words: **sun, earth, moon, and light**.

② A lunar eclipse happens only when the three big objects line up **exactly**: sun — earth — moon. On a normal full-moon night, the moon is lit directly by the sun. But during an eclipse, the earth stands in between. The earth's shadow falls on the moon, and the direct sunlight is **cut off**. You might expect the moon to become completely dark. And yet, strangely, it often looks **bright red**, clearly visible without a telescope. Why?

③ The answer lies in a clever trick of our **atmosphere** — the thin layer of air around the earth. When sunlight passes around the edge of the earth, it must travel through this air. Air is not empty: it is full of tiny particles of water, dust, and gas. These tiny particles **scatter** light of different colours in different amounts. **Blue light is scattered the most; red light is scattered the least.** This is the same reason our daytime sky looks blue and our sunsets look red.

④ During a lunar eclipse, then, only the **reddish** part of the sun's light is able to "bend around" the earth and continue on to the moon's surface. The moon acts like a great white screen, hanging far away in space. It catches this soft, red light and **throws it back** to our eyes. So the "blood moon" is not a sign of anything strange. **It is simply the sum of all the sunrises and sunsets on earth at that moment, gently painted on the face of the moon.** A poet once wrote: "*On a blood-moon night, the earth lights the moon with the red of all its quiet dawns.*"

⑤ The exact shade of red changes. Sometimes the moon is a dim **brown-orange**, like weak tea; sometimes it glows a strong **copper**; once, in 1883 after the great Krakatoa volcano, the moon was almost **black**. Volcanoes, dust storms and forest fires all push more tiny particles into the sky, so less red light can reach the moon. Astronomers even use a scale called the **Danjon Scale** (0 to 4) to measure the moon's colour during each eclipse: 0 means nearly invisible, 4 means a bright red-orange clearly seen by everyone.

⑥ A lunar eclipse is, as the American space agency NASA has repeated many times, **completely safe to watch**. Unlike a solar eclipse, where staring at the sun can damage

the eyes in seconds, the moon's light — even at full brightness — is far too weak to hurt us. A good pair of binoculars or a small telescope can show you the **craters** on the moon's edge, dark ring inside the "blood". Even bare eyes, on a windless evening, are enough.

⑦ A few simple tips for the next one. First, **look up the time**. A full eclipse lasts from about an hour to an hour and a half, but the deepest red usually appears for 10—20 minutes near the middle. Second, **find a dark place**. City lights wash out the gentle colour; even a small park is better than a balcony over a bright street. Third, **bring a blanket**. A late-autumn night sky is colder than you think, and a blood moon is something to watch **patiently**, not to glimpse in a hurry.

⑧ The next total lunar eclipse visible from most of China will be **on 31 December 2028**. If you get a chance, go outside with your family. Stand quietly and look up. You will see a soft red moon — and you will know that what you are really watching is the **light of all the day's goodbyes**, travelling around the earth to find the moon. Science, when it is clearly understood, does not take the beauty out of the night. It **puts it back in**.

注释： lunar eclipse 月食；shadow 影子；atmosphere 大气层；scatter 散射；copper 铜色；crater 环形山

根据短文内容，选择最佳答案。

1. Why does the moon NOT simply go dark during a lunar eclipse?

A. Because the moon produces its own weak red light.
B. Because the earth reflects sunlight back to the moon.
C. Because the earth's atmosphere scatters away most of the blue light but allows red light to bend around and reach the moon.
D. Because the sun comes out at night during an eclipse.

2. What IS the beautiful idea in Paragraph ④ about the colour on the moon?

A. The red colour is the result of a nearby volcano.
B. The red light is the sum of all the sunrises and sunsets on earth at that moment, painted on the moon's face.
C. The moon is burning inside.
D. Red dust in space is falling on the moon.

3. Why was the moon almost black in 1883?

A. Because scientists shut off the sun.
B. Because the great Krakatoa volcano had pushed a huge amount of tiny particles into the sky, so very little red light could reach the moon.

C. Because it was a solar eclipse.
D. Because of a dust storm on the moon.

4. Which of the following is TRUE about watching a lunar eclipse?

A. You must wear special eye-damage glasses.
B. You can only watch it with a large telescope.
C. It is completely safe to watch, because the moon's light is far too weak to harm eyes; even bare eyes are enough.
D. Looking at the moon during an eclipse damages the eyes in seconds.

5. According to Paragraph ⑦, which tip is NOT suggested?

A. Look up the time, since the deepest red usually appears for about 10—20 minutes.
B. Find a dark place away from city lights.
C. Bring a blanket, since the late-autumn night is cold.
D. Photograph the moon with a strong flashlight.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第③④段 "Blue light is scattered the most; red light ... scattered the least" + "reddish ... bend around the earth"
【解析】散射原理直引。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第④段 "the sum of all the sunrises and sunsets on earth at that moment, gently painted on the face of the moon"
【解析】核心比喻。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "1883 after the great Krakatoa volcano ... almost black"
【解析】火山粒子阻挡红光。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑥段 "completely safe to watch ... the moon's light ... far too weak to hurt us

... bare eyes ... are enough"
【解析】安全与否对比日食。A、B、D 均错。故选 C。

5.
【答案】D
【题型】细节理解题（否定）
【定位】第⑦段三条建议
【解析】A、B、C 均在文中。D 未被建议，且强闪光会影响观察。故选 D。